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Analysis Report
AI - CDR Biomarker Analysis
v1.0

**CDR Biomarker Reliability and the
Single-Threshold Bottleneck**

Project Subject: Glaucoma detection using intelligent
analysis of fundus images

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June 20, 2026

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1 Introduction

The end-to-end validation (companion report, see Section 7) showed that the pipeline is highly specific but only 54.8% sensitive. This report answers the decisive question behind that result: does the limited sensitivity originate in the AI segmentation, or in the diagnostic rule itself?

Using the same 267-image held-out validation set, the analysis proceeds in three steps: (i) it quantifies how faithfully the pipeline reproduces the area cup-to-disc ratio (aCDR), (ii) it benchmarks that agreement against the disagreement reported between human graders, and (iii) it tests whether even perfect, ground-truth masks would resolve the sensitivity problem.

2 aCDR Agreement: AI vs. Physician

Because the diagnostic decision depends entirely on the aCDR, the first question is how closely the aCDR computed from the predicted masks matches the aCDR derived from the physician ground-truth masks (Figure 1).

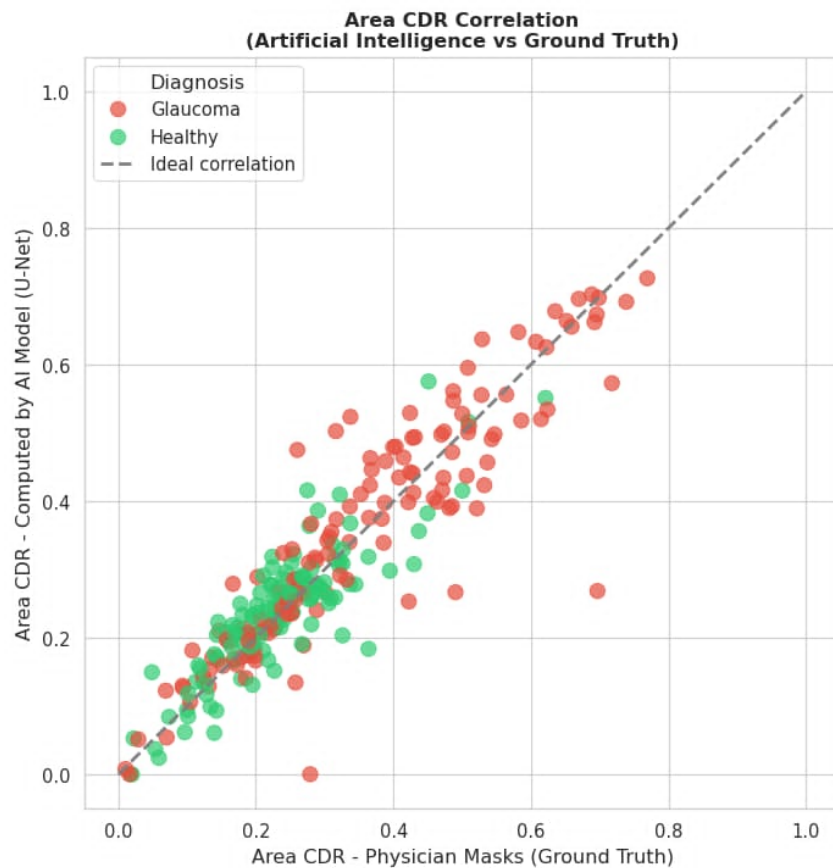


Figure 1: AI-derived area CDR versus physician ground-truth area CDR across the 267 validation images. Points cluster around the ideal diagonal.

Across the 267 images the agreement is strong: mean absolute aCDR error 0.044, median 0.031, RMSE 0.065, Pearson $r = 0.91$ ($R^2 = 0.83$), and a negligible mean signed error

of +0.003 (no systematic over- or under-estimation). The error distribution is strongly right-skewed-the median (0.031) lies well below the mean-so the average is inflated by a few hard cases rather than a broad bias: 92.5% of images are reproduced to within 0.10 aCDR of the expert value, only 7.5% exceed 0.10, just 1.5% exceed 0.20, and the worst single case reaches 0.428.

Conclusion. The segmentation stage reproduces the diagnostic biomarker faithfully; aCDR estimation is not a meaningful source of error.

3 Comparison with Human Inter-Observer Variability

This agreement is best judged against how much human graders themselves disagree. Clinical cup-to-disc grading is only moderately reproducible: Tielsch et al. (1988) report interobserver agreement of $\kappa = 0.71$ (horizontal) and $\kappa = 0.74$ (vertical) for the cup-to-disc ratio graded from stereo disc photographs. Evidence specific to the *area* CDR is scarcer, because clinicians conventionally estimate the ratio by diameter; where six ophthalmologists independently delineated the disc and cup *areas* in the RIGA dataset (Almazroa et al. 2018), the inter-observer spread was large enough that outlying annotations had to be filtered before the contours could serve as a reference standard.

Conclusion. Because these human benchmarks are partly diameter-based the comparison is indicative rather than exact; nonetheless, the pipeline’s aCDR error (0.044) and correlation ($r = 0.91$) are of the same order as-and in several respects better than-documented human disagreement. The automated biomarker is at least as reliable as a single human grader.

4 Separability and the Perfect-Mask Test

If neither the segmentation nor the biomarker estimate is at fault, the limitation must lie in the decision rule. Figure 2 shows the distribution of aCDR values for healthy and glaucomatous eyes, for both ground-truth and AI masks.

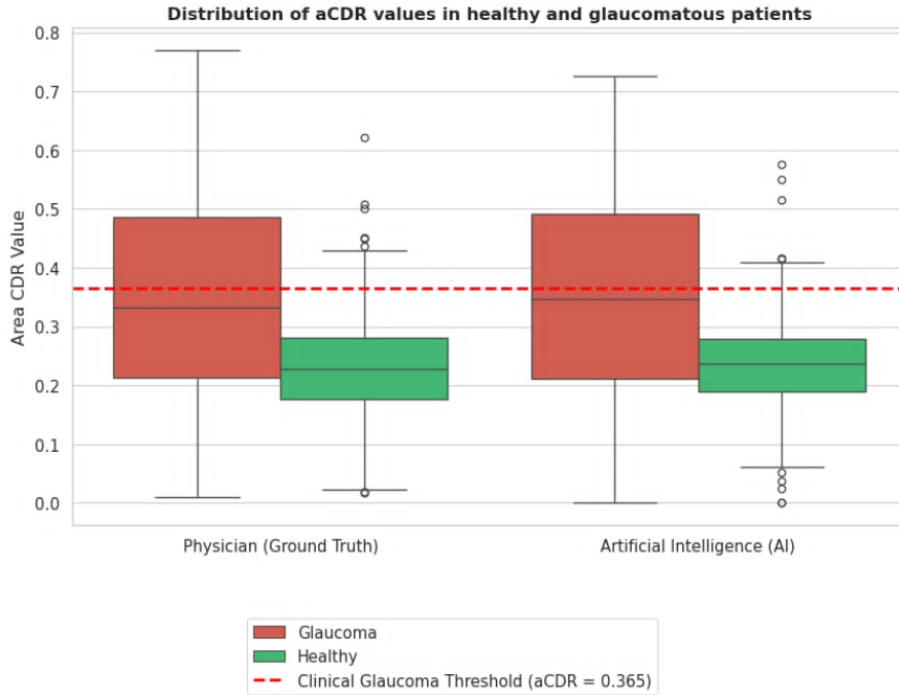


Figure 2: Distribution of aCDR for healthy vs. glaucomatous eyes (physician ground truth and AI predictions). The dashed line marks the clinical aCDR = 0.365.

The healthy and glaucomatous distributions *overlap substantially*, so no single cut-off can separate them cleanly. The decisive experiment is to apply the clinical area-CDR threshold ($\text{aCDR} \geq 0.365$) to the *ground-truth physician masks*-an idealised configuration equivalent to perfect segmentation. This yields only 45.9% sensitivity (accuracy 69.7%, specificity 93.9%, F1 60.5%), *lower* than the AI pipeline’s 54.8%, whose threshold is calibrated on the data.

Conclusion. Perfect masks do not fix the sensitivity problem. The bottleneck is therefore the univariate, single-threshold aCDR decision rule itself, not the segmentation network: many true glaucoma cases simply present with an aCDR below any useful threshold.

5 Implications and Recommended Direction

This finding redirects the optimisation effort. Improving segmentation quality further-the subject of the *Proposed Improvements to the U-Net++ Model* report-will raise mask accuracy but cannot, by itself, lift screening sensitivity past the ceiling imposed by a single biomarker. The two directions are complementary but distinct: that report targets the *masks* (Dice score), whereas the limitation identified here is in the *decision rule*.

The recommended direction is to replace the single aCDR threshold with a **multi-feature classifier** that combines the area CDR with additional, already-catalogued biomarkers-neuroretinal rim morphology, the ISNT rule, RADAR and DDLS, peripapillary atrophy, disc haemorrhages, and inter-eye asymmetry (inventoried in the *Input Features, Methods and Resources* report). Several of these-ISNT-rule violations in particular-flag exactly the early, focal cases that a global aCDR misses, and are therefore the most promising route

to raising sensitivity above the current ceiling.

6 Summary

- The pipeline reproduces the physician aCDR faithfully (mean error 0.044, $r = 0.91$, 92.5% within 0.10), comparable to or better than human inter-observer variability.
- The healthy/glaucoma aCDR distributions overlap, and the clinical threshold on *perfect* ground-truth masks gives only 45.9% sensitivity.
- The bottleneck is therefore the single-threshold aCDR rule, not the AI segmentation.
- Recommended direction: a multi-feature classifier (CDR + ISNT/NRR, RADAR, DDLS, atrophy, asymmetry), complementary to the ongoing segmentation-quality improvements.

7 Related Team Documents

- *End-to-End Validation of the Glaucoma Detection Pipeline* - companion report providing the classification metrics analysed here.
- *Input Features, Methods and Resources* - catalogue of the additional biomarkers proposed for the multi-feature classifier.
- *Proposed Improvements to the U-Net++ Model* - complementary effort targeting mask quality (Dice) rather than the decision rule.

8 References

1. Tielsch JM, Katz J, Quigley HA, Miller NR, Sommer A. Intraobserver and inter-observer agreement in measurement of optic disc characteristics. *Ophthalmology*. 1988;95(3):350-356.
2. Almazroa A, Alodhayb S, Osman E, et al. Retinal fundus images for glaucoma analysis: the RIGA dataset. *Proc. SPIE 10579, Medical Imaging 2018*; 105790B.